Chapter 11 - Matrix Inverse and Condition

$A\bar{A}^{-1} = \bar{A}^{-1}A = I$ $\quad$ A: square matrix

Calculate the Inverse

- Compute the inverse column-by-column fashion

if $b = \begin{bmatrix}1\\0\\0\end{bmatrix}$ $\longrightarrow$ solution to $Ax=b$ will give the first column of $\bar{A}^{-1}$.

$\rightarrow x = \bar{A}^{-1}b$.

if $b = \begin{bmatrix}0\\1\\0\end{bmatrix}$ $\rightarrow$ solution of $Ax=b$ will give the 2nd col. of $\bar{A}^{-1}$.

- Can use LU factorization to solve

1st col. of $\bar{A}^{-1}$: $\underbrace{LUx}_{d} = \begin{bmatrix}1\\0\\0\end{bmatrix} \Rightarrow Ld = \begin{bmatrix}1\\0\\0\end{bmatrix} \rightarrow d \Rightarrow Ux = d \rightarrow$ solve x (1st col. of $\bar{A}^{-1}$)

2nd col. of $\bar{A}^{-1}$: $\underbrace{LUx}_{d} = \begin{bmatrix}0\\1\\0\end{bmatrix} \Rightarrow Ld = \begin{bmatrix}0\\1\\0\end{bmatrix} \rightarrow d \Rightarrow Ux = d \rightarrow$ solve x (2nd col. of $\bar{A}^{-1}$)

3rd col. of $\bar{A}^{-1}$: $\underbrace{LUx}_{d} = \begin{bmatrix}0\\0\\1\end{bmatrix} \Rightarrow Ld = \begin{bmatrix}0\\0\\1\end{bmatrix} \rightarrow d \Rightarrow Ux = d \rightarrow$ solve x (3rd col. of $\bar{A}^{-1}$)

Error Analysis and System Condition

$$Ax = b.$$

if small change in A → small change in x → "well-conditioned"

" " A → large change in x → "ill-conditioned"

Ex

$$\overbrace{\begin{bmatrix} 400 & -201 \\ -800 & 401 \end{bmatrix}}^{A} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 200 \\ -200 \end{bmatrix} \Rightarrow \text{solution} \quad \begin{matrix} x_1 = -100 \\ x_2 = -200 \end{matrix}$$

Make a small change in A

$$\begin{bmatrix} 401 & -201 \\ -800 & 401 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 200 \\ -200 \end{bmatrix} \Rightarrow \text{solution:} \quad \begin{matrix} x_1 = 40000 \\ x_2 = 78800 \end{matrix}$$

very large change

"ill-conditioned"

- Inverse provides a means to understand if a system is ill-conditioned.
- Scale A s.t. largest element in each row is 1. Invert the scaled A. If some elements of A^{-1} are very large as compared to 1 ⇒ system is ill-conditioned

Vector and Matrix Norms:

n-dimensional vector $X = [x_1 \dots x_n]$

Euclidean norm: $\|X\|_e = \sqrt{\sum_{i=1}^{n} x_i^2}$

Matrix A: Frobenius norm: $\|A\|_f = \sqrt{\sum_{i=1}^{n} \sum_{j=1}^{n} a_{ij}^2}$

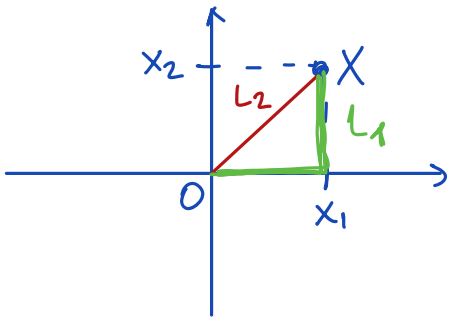
- A single number that quantifies "size" of A.

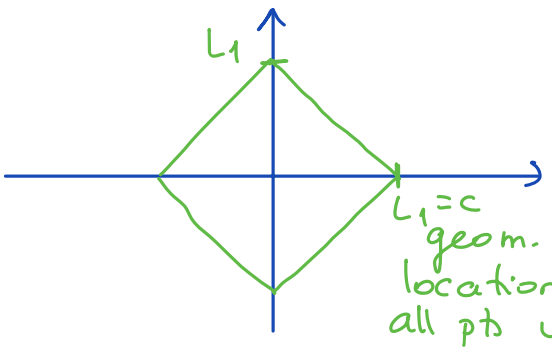
p-norms

$$\|X\|_p = \left(\sum_{i=1}^{n} |x_i|^p\right)^{1/p} \rightarrow \begin{cases} p=2 \Rightarrow \text{Euclidean norm } (L_2 \text{ norm}) \\ p=1 \Rightarrow L_1 \text{ norm.} \end{cases}$$

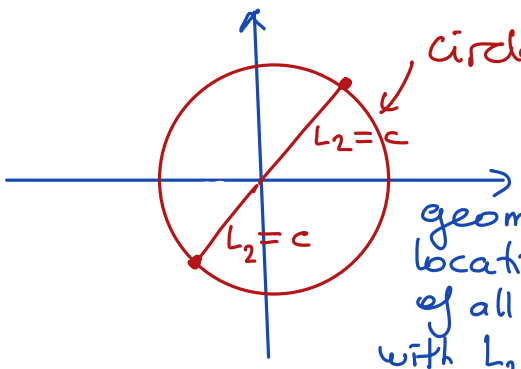
$$\|X\|_1 = \sum_{i=1}^{n} |x_i|$$

$$\|X\|_\infty = \max_{1 \le i \le n} |x_i|$$





geom. location of all pts with $L_1 = c$ norm

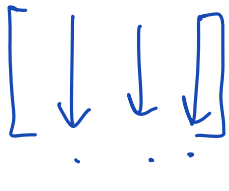


geom. location of all pts with L_2 norm.

Matrices:

$$\|A\|_1 = \max_{1 \le j \le n} \sum_{i=1}^{n} |a_{ij}|$$

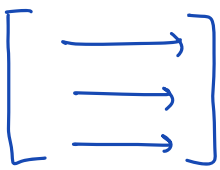
column-sum norm.



max

$$\|A\|_\infty = \max_{1 \le i \le n} \sum_{j=1}^{n} |a_{ij}|$$

row-sum norm



max

spectral norm

$$\|A\|_2 = (\mu_{max})^{1/2}$$

largest eigenvalue of $A^T A = \mu_{max}$

Matrix Condition Number

$\text{cond}(A) = \|A\|_\infty \|\bar{A}^{-1}\|_\infty$ ill-conditioned if $\text{cond}(A) \gg 1$.

EX $A = \begin{bmatrix} 400 & -201 \\ -800 & 401 \end{bmatrix} \Rightarrow \|A\|_\infty = \max\{(|400|+|-201|), (|-800|+|401|)\}$
$= 1201$

$$\bar{A}^{-1} = \frac{1}{(400)(401)-(800)(201)} \begin{bmatrix} 401 & 201 \\ 800 & 400 \end{bmatrix} = \begin{bmatrix} -1.002 & -0.502 \\ -2 & -1 \end{bmatrix}$$

$\|\bar{A}^{-1}\|_\infty = \max\{(|-1.002|+|-0.502|), (|-2|+|-1|)\} = 3$

$\text{Cond}(A) = 1201 \cdot 3 = 3603 \gg 1 \Rightarrow$ ill-conditioned.

EX $A = \begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{3} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} \end{bmatrix}$

- 3×3 Hilbert matrix
- Use row-sum norm to estimate the condition number

- Normalize each row s.t. max element in each row is 1

$$A = \begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{3} \\ 1 & \frac{2}{3} & \frac{1}{2} \\ 1 & \frac{3}{4} & \frac{3}{5} \end{bmatrix} \rightarrow \|A\|_\infty = 1 + \frac{3}{4} + \frac{3}{5} = 2.35 \quad (3^{rd} \text{ row})$$

$$\bar{A}^{-1} = \begin{bmatrix} 9 & -18 & 10 \\ -36 & 96 & -60 \\ 30 & -90 & 60 \end{bmatrix} \rightarrow \|\bar{A}^{-1}\|_\infty = |-36| + 96 + |-60| = 192 \quad (2^{nd} \text{ row sum})$$

$\text{Cond}(A) = (2.35)(192) = 451.2 \gg 1$

$\Rightarrow$ "ill-conditioned"

MATLAB : `>> norm(X,p)` % p : (1, 2, inf, 'fro')

`>> cond(X,p)` % condition number

Interpreting the elements of matrix inverse as a measure of ill-conditioning.

- Scale A s.t. largest element in each row is 1.
- Invert the scaled A. If some elements of A^{-1} are very large as compared to 1 ⇒ system is ill-conditioned

insight $Ax = b$ we want to solve this.

$\hat{x}$: approximate solution

check if $\hat{x}$ is acceptable:

$$R = b - A\hat{x} \quad (1)$$

↳ residual vector

if R is small ⇒ $\hat{x}$ is acceptable

However, if X is the exact solution:

$$0 = b - AX \quad (2)$$

subtract (2) from (1):

$$R = A(X - \hat{X})$$

$$A^{-1}R = X - \hat{X}$$

↳ if elements of A^{-1} are large, a small R may give a large error $X - \hat{X}$.

⇒ A small residual does not guarantee an accurate sol^n.

→ if largest element of $A^{-1} \approx 1$ ⇒ system is well-conditioned

→ " " $\gg 1$ ⇒ " " ill-conditioned.

Other way to find A^{-1}

Gauss-Jordan elimination

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & -1 & 2 \\ 0 & 1 & 0 & -1 & 0 & 1 \\ 0 & 0 & 1 & 2 & 1 & -2 \end{array}\right]$$

A $\qquad$ A^{-1}

put left side into reduced row-echelon form. (make it an identity matrix)

Using Matrix of Minors and Cofactor matrix

$$C = \begin{bmatrix} -1 & -2 & 2 \\ 2 & 1 & 1 \\ 3 & 4 & 5 \end{bmatrix} \longrightarrow \begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

$(-1)^{i+j}$ $\quad$ i : row, j : col.

Matrix of minors

$$\begin{bmatrix} \begin{vmatrix} 1 & 1 \\ 4 & 5 \end{vmatrix} & \begin{vmatrix} 2 & 1 \\ 3 & 5 \end{vmatrix} & \begin{vmatrix} 2 & 1 \\ 3 & 4 \end{vmatrix} \\ \begin{vmatrix} -2 & 2 \\ 4 & 5 \end{vmatrix} & \begin{vmatrix} -1 & 2 \\ 3 & 5 \end{vmatrix} & \begin{vmatrix} -1 & -2 \\ 3 & 4 \end{vmatrix} \\ \begin{vmatrix} -2 & 2 \\ 1 & 1 \end{vmatrix} & \begin{vmatrix} -1 & 2 \\ 2 & 1 \end{vmatrix} & \begin{vmatrix} -1 & -2 \\ 2 & 1 \end{vmatrix} \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 7 & 5 \\ -18 & -11 & 2 \\ -4 & -5 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -7 & 5 \\ 18 & -11 & -2 \\ -4 & 5 & 3 \end{bmatrix}$$

cofactor matrix

$$\det(C) = \begin{vmatrix} -1 & -2 & 2 \\ 2 & 1 & 1 \\ 3 & 4 & 5 \end{vmatrix} \begin{matrix} -1 & -2 \\ 2 & 1 \\ 3 & 4 \end{matrix}$$

$6 \quad -4 \quad -20 \quad -5 \quad -6 \quad 16$

$$= -5 - 6 + 16 - (-20 - 4 + 6) = \underline{\underline{23}}$$

$$C^{-1} = \frac{1}{\det(C)} \underbrace{\text{adj}(C)}_{\text{transpose of cofactor}} = \frac{1}{23} \begin{bmatrix} 1 & 18 & -4 \\ -7 & -11 & 5 \\ 5 & -2 & 3 \end{bmatrix}$$